DoRAemon's Secret Tool for an A+ in Deep Learning

Weight-Decomposed Fine-Tuning

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Introduction

Full fine-tuning large language models is expensive. **LoRA** made fine-tuning cheaper by training low-rank updates while freezing the base model. **DoRA** makes this process even cheaper.

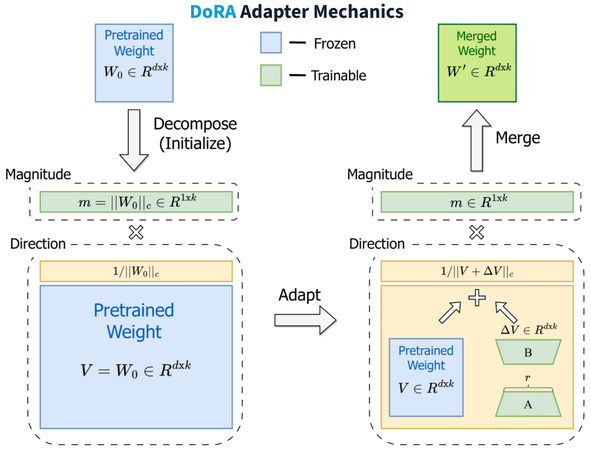
DoRA (Weight-Decomposed Low-Rank Adaptation)

- **DoRA** decomposes pretrained weights into **magnitude** and **direction**.
- Direction receives a **LoRA**-style low-rank update; magnitude is learned separately.

Project goal

- Maximize fine-tuning performance with **DoRA** while using fewer parameters.
- Test whether **DoRA** beats **LoRA** at the same rank and where it helps most: attention, MLP, or full-scope adapters.

DoRA Adapter Mechanics



Methodology

- Trained on 15k examples instead of 170k due to compute constraints.
- We otherwise followed the **DoRA** paper setup with a frozen base model, adapter-only checkpoints, 3 epochs, cutoff length 256, and seed 42.
- Settings: **LoRA** used rank 32 at LR=3e-4; rank-halved **DoRA** used rank 16 at LR=2e-4.

Our Independent Experiment: Scope Ablation

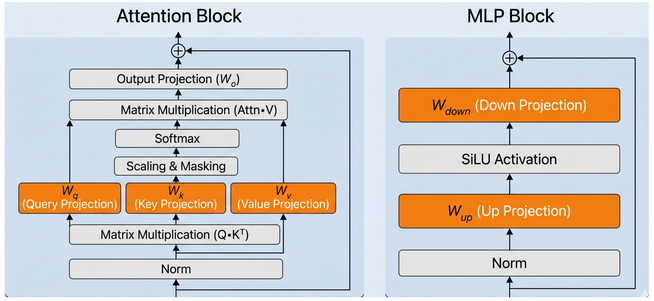


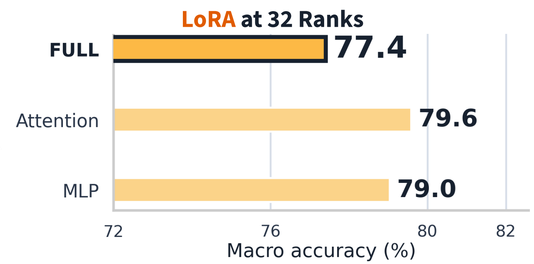
Figure 1. The ablation isolates adapter placement in attention projections, MLP projections, or both.

The ablation compares full (`q,k,v,up,down`), attention-only (`q,k,v`), and MLP-only (`up,down`) adapters to test whether DoRA works best when applied broadly or within a single layer group.

Verdict: DoRA Improves Accuracy and Efficiency

Accuracy	77.41%	vs	79.20%	+1.79	↑
Trainable Parameters	0.83%	vs	0.43%	-48.2%	↓

The Adapter Duel



LoRAx vs. **DoRAemon**

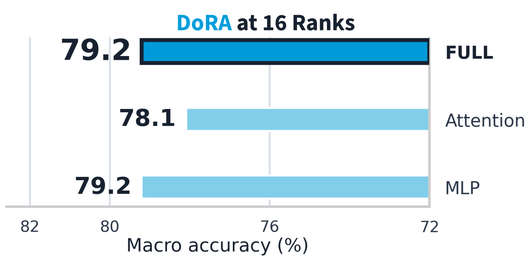


Figure 2. The full-scope bars are outlined: rank-halved **DoRA** improves macro accuracy from 77.4% to 79.2%.

Benchmark Example

HellaSwag — Grounded Activity Completion (+6.0 pts)

Given a short activity description, select the most plausible four-way continuation. This benchmark produced the largest full-scope **DoRA** gain, improving by **+6.0 points**. Requires grounded physical-world commonsense.

Context: A woman is outside with a bucket and a dog. The dog is running around trying to avoid a bath. She pours soap on the dog and…

(A) lathers the dog up with a brush before rinsing.
(B) pours more water slowly into the bucket.
(C) scrubs the dog down and rinses it thoroughly.
(D) lets the dog run off and air dry on the lawn.

Official Paper vs Full-Scope LLaMA2-7B Reproduction

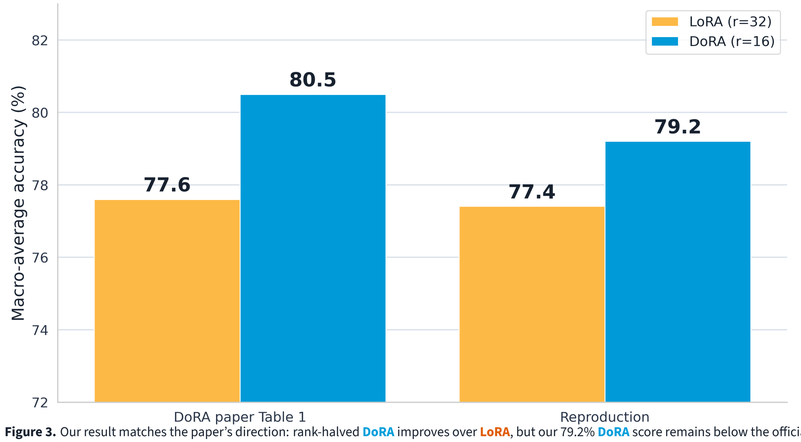


Figure 3. Our result matches the paper's direction: rank-halved **DoRA** improves over **LoRA**, but our 79.2% **DoRA** score remains below the official 80.5% reference.

LoRA vs DoRA Across Ranks

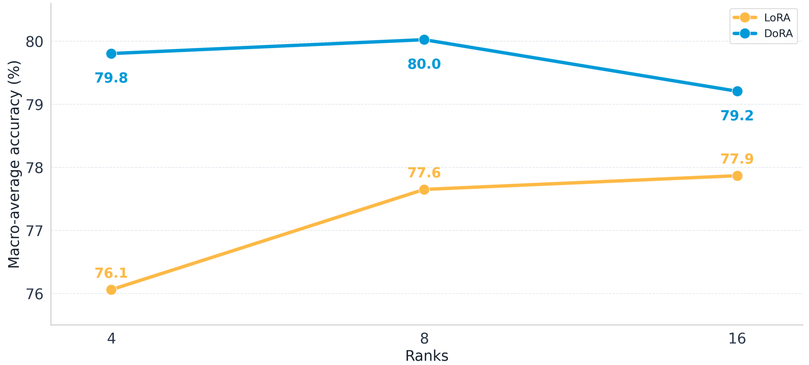


Figure 4. Macro-average accuracy across rank sizes, with the DoRA advantage shown in the lower panel.

Conclusion & Future Work

Takeaways

- Full-scope **DoRA** improved macro-average accuracy by **+1.79 points** while using fewer trainable parameters.
- **LoRA** is still strong at higher ranks, but still performs worse than **DoRA**.
- **DoRA** worked best when applied fully across the transformer.

Potential Next steps

- Train on the full `commonsense_170k` dataset to address small dataset worries.
- Repeat the LoRA and DoRA variants across multiple random seeds to verify consistency.
- Check more rank comparisons at full scale to verify the trend across more ranks.

References

[1] Edward J. Hu et al. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

[2] Hugging Face. Transformers documentation. `https://huggingface.co/docs/transformers`, 2026.

[3] Shih-Yang Liu et al. DoRA: Weight-decomposed low-rank adaptation. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. Oral presentation.

[4] NVIDIA Research. NVlabs/DoRA: Official implementation. `https://github.com/NVlabs/DoRA`, 2024.